

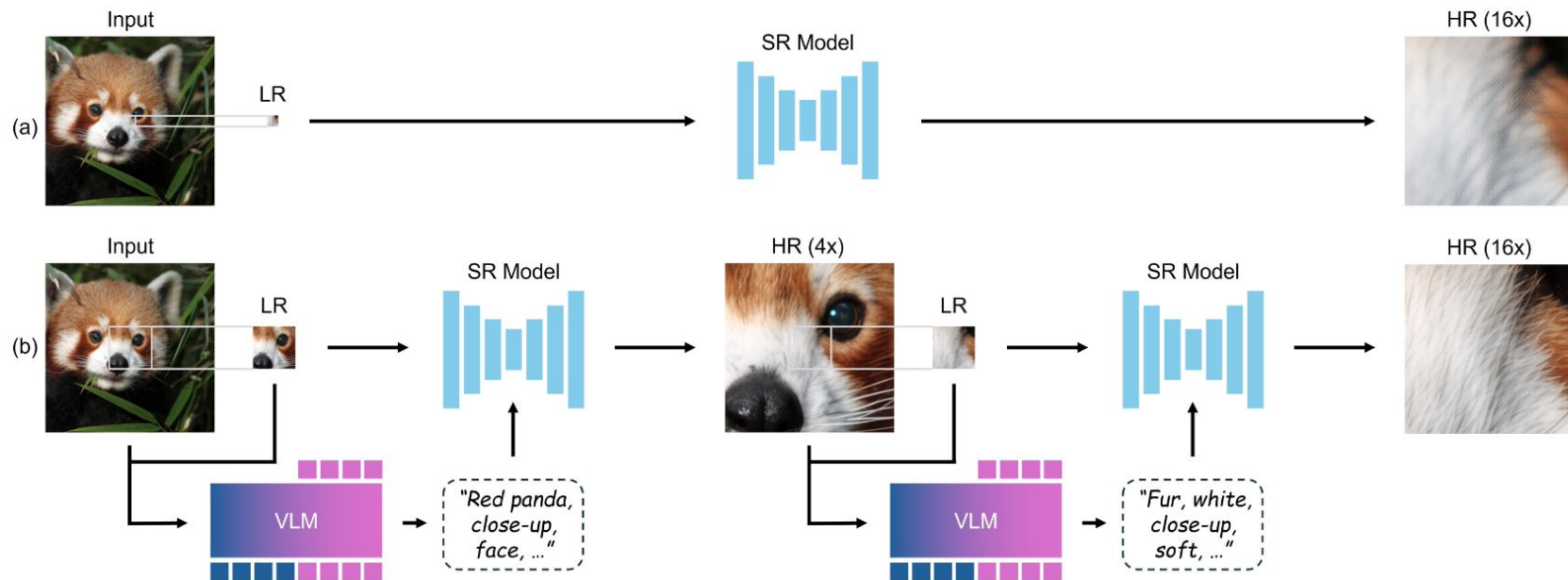
Final Project Proposal

group 6

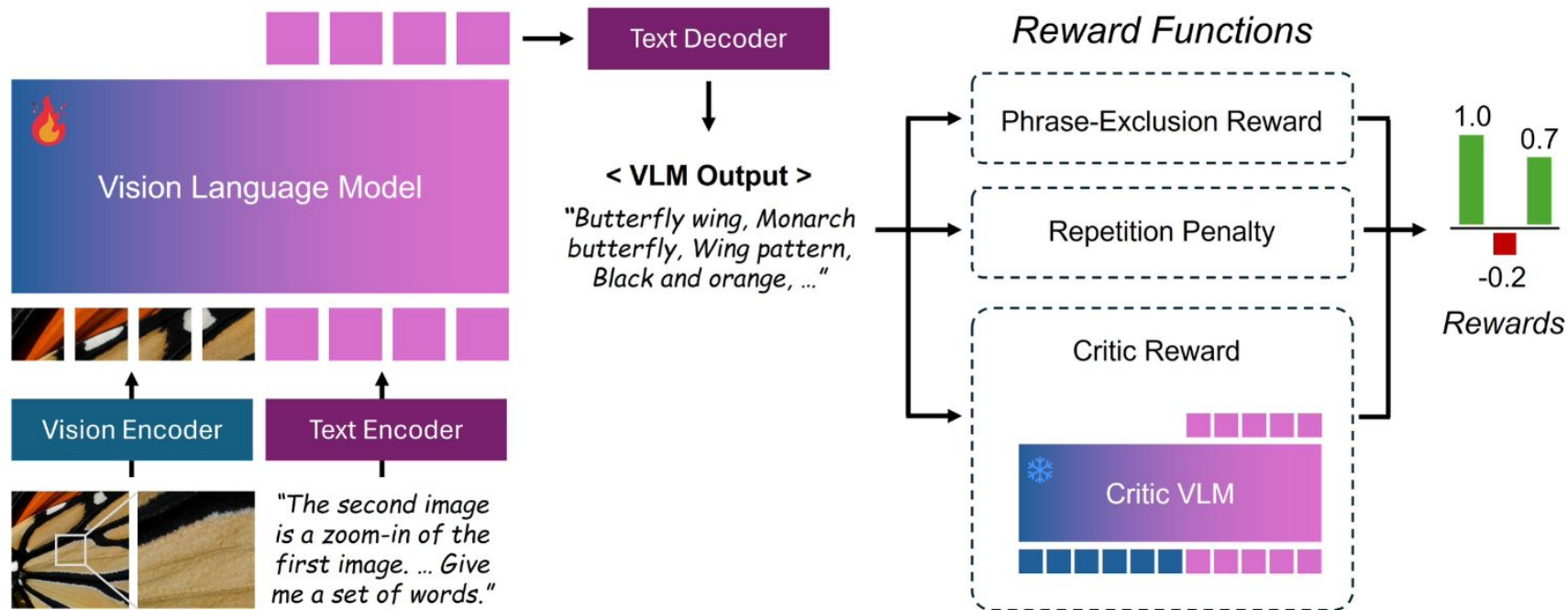
Chain-of-Zoom (NeurIPS 2025 Spotlight)



Pipeline



GRPO Training Framework

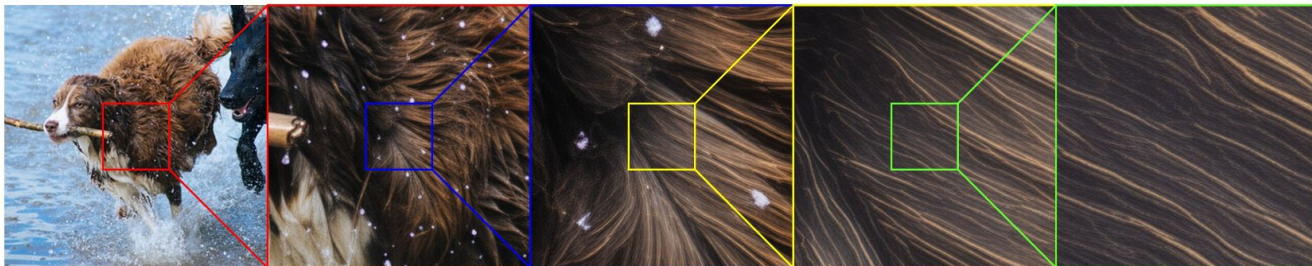


Reward

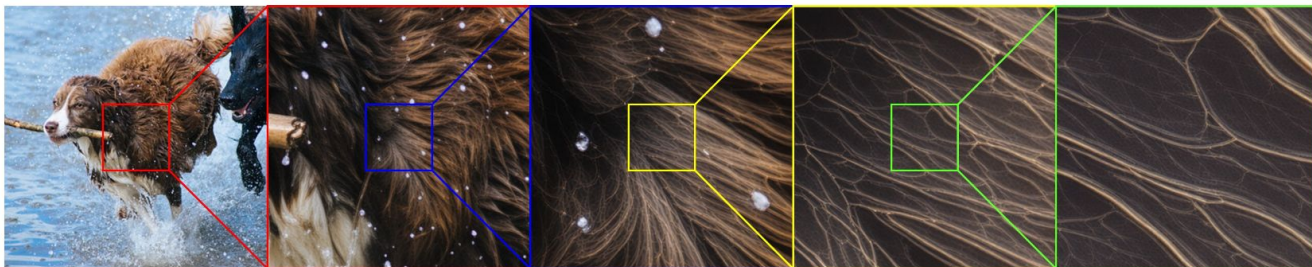
- Critic Preference Reward
 - Critic VLM (InternVL2.5-8B) judges the prompt of the input multi-scale image crops
- Phrase-Exclusion Reward
 - Predefined blacklist of unwanted phrases such as “first image” or “second image”
- Repetition Penalty
 - Negative reward for a higher repetition ratio

$$R(\mathbf{c}_i) = w_{\text{critic}} R_{\text{critic}} + w_{\text{phrase}} R_{\text{phrase}} + w_{\text{rep}} R_{\text{rep}}$$

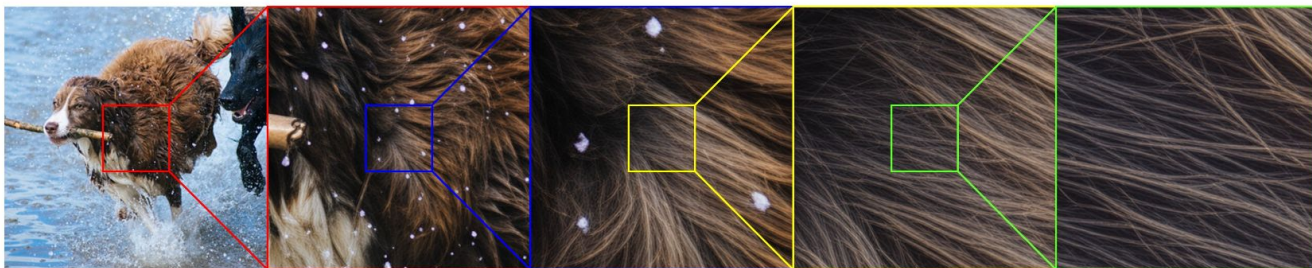
$p_\phi(\mathbf{c}_i \mid \mathbf{x}_{i-1})$



$p_\phi(\mathbf{c}_i \mid \mathbf{x}_{i-1}, \mathbf{x}_{i-2})$



+ GRPO Training



Proposal 1 - Resolution Enhancement

In the current recursive multiscale mode, the Vision-Language Model (VLM) operates with limited contextual awareness

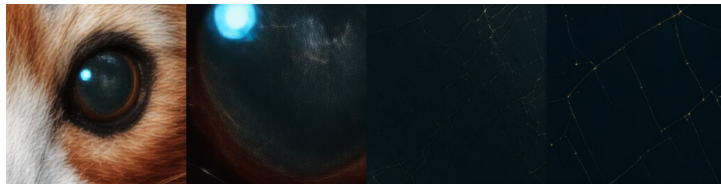
- Loss of Historical Context
- Scale Disorientation
- Semantic Drift

1. fur, texture, close-up, animal, soft, fluffy, white, brown, detail, macro, natural, fur texture, animal fur, closeup,
2. hair texture, fur, close-up, natural, soft, detailed, pattern, strands, beige, brown, white, realistic, naturalistic, close-up view



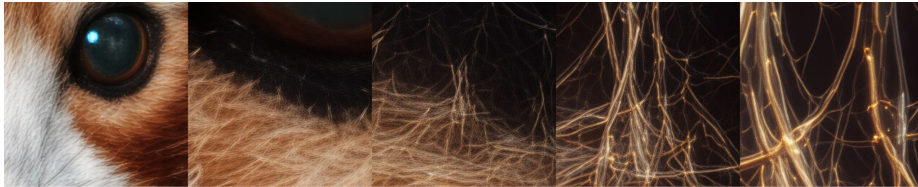
Semantic Drift

1. **dog eye**, close-up, fur, blue, brown, animal, eyes, detail, texture, whiskers, whisker, whiskers, whiskery,
2. darkness, blue light, **spider web**, close-up, texture, night, moonlight, macro, intricate, webbing, dark sky, glowing, spider
3. dark, spider web, close-up, night, texture, intricate, webbing, dark background, spider, web, macro, details, pattern, night sky



Semantic Drift

1. **dog eye**, close-up, **fur**, blue, brown, detail, texture, animal, eyes, macro, photography, pet, eyesight, whiskers,
2. **hair**, close-up, texture, detail, fur, animal, skin, macro, natural, brown, black, soft, fine, close, zoomed,
3. **Neurons**, dendrites, axons, synapses, neural network, brain activity, electrical signals, neurotransmitters, synaptic transmission, neural plasticity, learning
4. **Neurons**, dendrites, axons, synapses, neural network, brain activity, electrical impulses, neurotransmitters, synaptic transmission, neural signaling, complex structure



Solution

- State Expansion
 - Include information of previous prompt and scale factor
- Scale Progress / Information Gain
 - Penalty on word overlap with previous prompt
- Semantic Anchor Reward
 - Use the label of the original image as semantic anchor